# An Intro to Natural Language Processing

GENERAL ASSEMBLY

# Agenda

- What is NLP and what is it good for?
- Some NLP vocabulary
- Text Preprocessing
- Sentiment analysis

# What is Natural Language?

*"To understand text analytics and natural language processing, we need to understand what makes a language "natural." In simple terms,* ***natural language*** *is one developed and evolved by humans through natural use and communication, rather than constructed and created artificially, like a computer programming language."*

- Dipanjan Sarkar, *Text Analytics with Python: A Practitioner's Guide to NLP*

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# What is Natural Language Processing?

**Natural Language Processing (NLP)** is a specialized field of computer science primarily concerned with designing and building applications and systems for facilitating the interaction between machines and the natural language of humans.

*[Definition paraphrased from Sarkar.]*

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# What is NLP used for?

NLP has a several applications, including:

- **Machine translation**: Providing a systematic, grammatically correct translation between two natural languages
- **Speech recognition**: *“Siri, play Despacito.”*
- **Question and Answering Systems**: *“Siri, what is the capital of France?”*
- **Text summarization**: Take a piece of text and condense it to a much smaller version that still contains the basic information
    - Sentiment analysis is a type of this!
- **Text categorization**: Given a document, does it belong to category A or B?
    - e.g. Is this email *spam* or *ham*?
    - You’ll do this in the next lesson!
- **Text clustering**: Grouping documents based on their overall vibe
    - Outside the scope of our class… but only barely!

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# Some NLP Vocab

# NLP Vocab I

- A **document** is a piece of text that represents one item of analysis
  - For example, this could be a Yelp review, a Tweet, or a news article

- A group of documents is called a **corpus**. You can think of this as like a dataset of documents. The plural of corpus is **corpora**.

- A document is further broken down into **tokens**, the basic unit of analysis we're interested in. Generally, tokens are just **words**. The act of converting a document into tokens is **tokenizing**.

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# Tokens

For most NLP tasks, **tokens** are just **words**.

Tokens can be smaller than words, such as **characters** or **syllables** for some tasks, such as text-to-speech.

In our class, we'll also see ***n*-grams** as tokens

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# *n*-grams

An ***n*-gram** is a tuple of *n* words used as a whole token. For example:

| | |
|---|---|
| **original document** | the cat sat on the mat |
| **unigrams (words)** | the, cat, sat, on, the, mat |
| **bigrams** | (the, cat), (cat, sat), (sat, on), (on, the), (the, mat) |
| **trigrams** | (the, cat, sat), (cat, sat, on), (sat, on, the), (on, the, mat) |

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# NLP Vocab II

Traditional, tabular data:

*“We’re going to **preprocess** our **dataset**.”*

Text data:

*“We’re going to do **text normalization** on our **corpus**.”*

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# Text Normalization

The **normalization** (ie, preprocessing) of your corpus could involve the following steps:

- Tokenization
- Removing or replacing special characters and punctuation ($#@%&, etc)
- Expanding contractions (don't ↦ do not)
- Removing **stop words**
- Correcting repeating characters/typos (heyyy ↦ hey)
- **Stemming** our tokens
- **Lemmatizing** our tokens
- **Part of Speech Tagging**

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# Stop Words

In English, **stop words** are small words that have little or no significance to the meaning of the overall sentence.

| | |
|---|---|
| Original Sentence | *I am eating a sandwich from the restaurant by the store* |
| After stop word removal | *eating sandwich restaurant store* |
| | |

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# Stemming

To **stem** a word, we remove its suffixes so that multiple forms of the words are recognized by their root.

- jumps ↦ jump
- jumped ↦ jump
- jumping ↦ jump

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# Stemming

Here's our example using the most popular **Porter stemmer**. Note that the stemmer is algorithmic and not perfect.

| | |
|---|---|
| Original Sentence | *I am eating a sandwich from the restaurant by the store* |
| After stop word removal | *eating sandwich restaurant store* |
| After stemming | *eat sandwich restaur store* |

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# Lemmatizing

Similar to stemming, **lemmatizing** reduces a word to its base form. It's a lot smarter in that it can conjugate verbs and has more rules, but is therefore much slower.

- going ↦ go
- went ↦ go
- gone ↦ go

# Lemmatizing

Another major issue with lemmatizing is that you need to know a word's **part of speech** or else the algorithm won't know how to conjugate

- wound ↦ wind (as a verb)
- wound ↦ wound (as a noun)

# Lemmatizing

Here's our example again using the popular **WordNet lemmatizer**.

| | |
|---|---|
| Original Sentence | *I am eating a sandwich from the restaurant by the store* |
| After stop word removal | *eating sandwich restaurant store* |
| After lemmatizing | *eat sandwich restaurant store* |

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# Aside: Bag-of-words

It turns out, the order of words in a sentence (as dictated by English grammar) isn't hugely impactful to our understanding of a sentence!

Many NLP models are **bag-of-words models**. That is, just like words in a bag, the models won't take word order into account!

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# Bag-of-words examples

What is the overall sentiment of this *unordered, normalized* sentence?

*french omelette restaurant disgusting*

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# Bag-of-words examples

What is the overall sentiment of this *unordered, normalized* sentence?

*french omelette restaurant disgusting*

It was actually:

*The omelette at this French restaurant was disgusting!*

# Bag-of-words examples

How about another:

*quiche spinach not try bad*

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# Bag-of-words examples

How about another:

*quiche spinach not try bad*

It was actually:

*Try the spinach quiche, it was not bad!*

# Bag-of-words examples

How about another:

*quiche spinach not try bad*

It was actually:

*Try the spinach quiche, it was not bad!*

but it also could have been...

*Do not try the spinach quiche, it was bad!*

# Sentiment Analysis

# Sentiment Analysis

A common and simple NLP task is **sentiment analysis**. That is, assigning a sentiment value to a given document based on how positive or negative it is. For example:

**Negative sentiment document:** *This was a terrible movie, it was so bad I wanted to vomit!*

**Positive sentiment document:** *Our server was very courteous, and the meal was delicious!*

**Neutral sentiment document:** *I have tickets for the 2pm train.*

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# Sentiment Analyzers

Most sentiment analysis models are bag-of-words models that assign point values to individual words. Here are those examples passed through the VADER lexicon sentiment analyzer. VADER was fit to work best with social media text.

| Document | VADER Lexicon Sentiment |
|---|---|
| *This was a terrible movie, it was so bad I wanted to vomit!* | `{'neg': 0.458, 'neu': 0.542, 'pos': 0.0, 'compound': -0.822}` |
| *Our server was very courteous, and the meal was delicious!* | `{'neg': 0.0, 'neu': 0.513, 'pos': 0.487, 'compound': 0.8217}` |
| *I have tickets for the 2pm train.* | `{'neg': 0.0, 'neu': 1.0, 'pos': 0.0, 'compound': 0.0}` |

# Python Time!

